# AJAN MUTHURAJ

**Robotics Software & Autonomy | IIT Madras** | Portfolio | GitHub | LinkedIn | ajan.muthuraj@gmail.com

## RESEARCH AND COMPETITION HIGHLIGHTS

- **Multi-Robot Frontier-Based Exploration under Uncertainty with SLAM and Temporal Map Fusion** · Co-author, ICRA 2026 XPLORE workshop poster.
- **2nd among Indian teams**, Swarm Rescue Challenge 2025–26 · Built a ten-drone search-and-rescue system with the IIT Madras team.
- **6th place**, e-Yantra Robotics Competition 2024–25, Logistic coBot · Validated UR5/eBot package handling at IIT Bombay and was selected for eYSIP.
- **Top 10**, e-Yantra Robotics Competition 2023–24, GeoGuide · Built a working vision-guided rover that routed to events in priority order.

## EDUCATION

| PROGRAM | INSTITUTION | CGPA / % | YEAR |
|---|---|---|---|
| Mechanical Engineering (B.Tech) | Indian Institute of Technology Madras | **8.52/10** | **2026** |
| Class XII (CBSE) | Bharatiya Vidya Bhavan's V. M. Public School | **95.4%** | **2021** |
| Class X (CBSE) | Bharatiya Vidya Bhavan's V. M. Public School | **97.2%** | **2019** |

## RELEVANT SKILLS

**Robotics:** ROS 2, SLAM, Nav2, MoveIt 2, motion planning, RGB-D perception, state estimation, feedback control and reinforcement learning.
**Software and Tools:** C++, Python, MATLAB, Linux, Git, CMake, Docker Compose, OpenCV, PyTorch, Gazebo, RViz, Isaac Sim and ArduPilot.
**Mechatronics:** SolidWorks, Fusion 360, Arduino, ESP32, sensor integration and motor control. **Languages:** English, Hindi, Tamil, Japanese (basic).

## PROFESSIONAL EXPERIENCE

| | |
|---|---|
| **Systems Engineer**<br>**DLTX, Japan**<br>**Oct 2026 - Present** | • Reverse-engineered **Lotus Notes** applications from XML exports to identify form structure, data fields and legacy screen behavior.<br>• Mapped legacy form definitions into reusable **React components** and built a browser interface from the supplied XML exports.<br>• Built working knowledge of the **full stack**, from React and Java/Spring Boot through data persistence, Docker and Kubernetes. |
| **Humanoid Robot**<br>**IIT Roorkee · Intern**<br>**Jun - Jul 2025** | • Converted SolidWorks models to URDF and integrated **Nav2**, SLAM and differential-drive control for humanoid navigation in Gazebo.<br>• Linked **RGB-D perception** and 3D object localization to MoveIt 2 and OctoMap so the robot could approach and grasp around obstacles.<br>• Recorded **96.6% grasp success** across 30 simulated trials and documented navigation and manipulation in separate demonstrations. |

## PROJECTS

### EXTRACURRICULAR PROJECTS

| | |
|---|---|
| **Isaac Sim UAV Autonomy** | • Connected Isaac Sim physics to **ArduPilot SITL** and brought RGB, IMU, range and LiDAR data into ROS 2 Jazzy via sensor topics and TF.<br>• Built **collision-aware 3D frontier exploration** with RTAB-Map and OctoMap, including replanning, return-to-home and landing.<br>• Ran **nine drones** in a shared Isaac Sim scene and published Docker Compose setup, flight-controller profiles and RViz telemetry.<br>• Fixed stale physics samples in the ArduPilot bridge and measured **5 cm peak horizontal error** over 61.7 s of simulated hover. |
| **Time-Varying Reach-Avoid RL**<br>**B.Tech Thesis** | • Added time to the **reach–avoid Bellman state** so a policy could react to moving obstacles and target zones in custom environments.<br>• Trained a **Double DQN** with experience replay and target-network updates to navigate under time-varying safety constraints.<br>• Measured **94.4% precision** among states predicted safe in the two-obstacle simulation and documented the result in the thesis poster. |
| **Swarm Rescue Challenge 2026** | • Contributed adaptive frontier selection and **temporal occupancy-grid fusion** to coordinate exploration across ten simulated UAVs.<br>• Contributed to **FastSLAM/ICP localization** and integrated hazard detection and collision recovery for GPS-denied search.<br>• Built for **single-operator supervision**, using shared maps to track fleet exploration during simulated search-and-rescue missions. |
| **Logistic coBot**<br>**e-Yantra, IIT Bombay** | • Integrated a **UR5 arm and eBot rover** in ROS 2 to pick up, sort and deliver packages across a shared logistics workspace.<br>• Localized packages with **ArUco** for MoveIt 2 Servo pick-and-place and fused rover measurements with an EKF for ultrasonic docking.<br>• Validated the complete **docking-to-pickup sequence** in Gazebo and on physical UR5 and eBot hardware at IIT Bombay. |
| **Cubitect**<br>**Electronics Club**<br>**Project Lead** | • Led a modular robotics project exploring self-reconfiguring cubes that use reaction wheels and controlled toppling to move.<br>• Built vision-based tracking with **A* path planning** and Hungarian assignment to match individual cubes with target positions.<br>• Developed the actuation and coordination approach and worked on a ROS 2 and Gazebo environment to test multi-cube behavior. |
| **Swarm Navigation**<br>**Inter-IIT Tech Meet** | • Developed a ROS 2 and Gazebo simulation workflow for groups of mobile robots navigating a dynamic warehouse-like environment.<br>• Combined LiDAR obstacle clustering with **DBSCAN**, map merging and TD3 policy training for multi-robot navigation.<br>• Published recordings, checkpoints and four- and five-robot launch configurations with the IIT Madras Inter-IIT robotics team. |
| **GeoGuide**<br>**e-Yantra, IIT Bombay** | • Built a physical mobile robot that used an overhead camera to localize arena events and navigate to them in priority order.<br>• Combined **ResNet-50** event classification and Dijkstra routing with ArUco tracking and QGIS georeferencing for the arena.<br>• Sent route commands to an ESP32 rover over Wi-Fi/TCP and used infrared sensors for on-robot line following during execution. |

### ACADEMIC PROJECTS

| | |
|---|---|
| **Decentralized Swarm Navigation** | • Extended **decentralized flocking** in MATLAB with shared occupancy-grid memory so robots could navigate an unknown maze.<br>• Implemented **deadlock recovery** to escape traps and delayed self-reinforcement to damp oscillations under noisy LiDAR sensing. |
| **Adaptive RRT* Motion Planning** | • Compared Adaptive RRT*, RRT*, A*, Adaptive A* and Lifelong Planning A* in simulated environments with dynamic obstacles.<br>• Implemented **incremental replanning** that keeps valid route segments and searches from a safe waypoint when obstacles move.<br>• Evaluated planners on path efficiency, robustness and computation, with source code and a static replanning diagram linked. |
| **Dual-Axis Solar Panel Tracker** | • Designed a dual-axis panel mechanism in Fusion 360 and used light-dependent resistors to sense the direction of incident light.<br>• Implemented **PID feedback** for two servos and added an OLED display for live voltage and status monitoring. |

## POSITIONS OF RESPONSIBILITY

| | |
|---|---|
| **Electronics Club**<br>**Core** | • Managed **recruitment, funding and inventory** for the IIT Madras Electronics Club's student projects and technical workshops.<br>• Taught a hands-on **parallel processing** workshop at Shaastra and helped run an IoT smartwatch hackathon from design to build. |
| **Electronics & Robotics Guild**<br>**Head** | • Led **electronics and robotics training**, selected members and assembled teams for Inter-IIT technical competitions.<br>• Represented IIT Madras in the **Bharat Forge swarm robotics challenge** and coordinated resources for student engineering teams. |

## EXTRA-CURRICULAR

- Captained Alakananda Hostel to **two IIT Madras football titles** and represented the institute at intercollegiate sports festivals.
- Represented IIT Madras in BGMI as part of the institute's gaming contingent and placed first in the intra-IIT tournament during the 2023-24 season.